MSE 480 Lab 3

Milling Procedure

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1. Introduction

This report outlines the milling procedure for machining a specified part from a 2" x 2" x 1.5" 6061 Aluminum rod as part of the Machining Lab #3 (MSE 480/780). The objectives of this lab are to identify milling parameters, understand milling limitations, differentiate between roughing and finishing operations, and simulate the milling process using Fusion 360. The deliverables include this report and a video submission of the milling simulation. The procedure is designed to maximize producibility by minimizing waste material, reducing machining time, and ensuring maximum accuracy, while achieving a surface finish of approximately 32 µinch. Additionally, this report evaluates the limitations of producing the part via additive manufacturing as an alternative to milling.

2. CAD Model

The part to be machined is detailed in an attached CAD drawing.

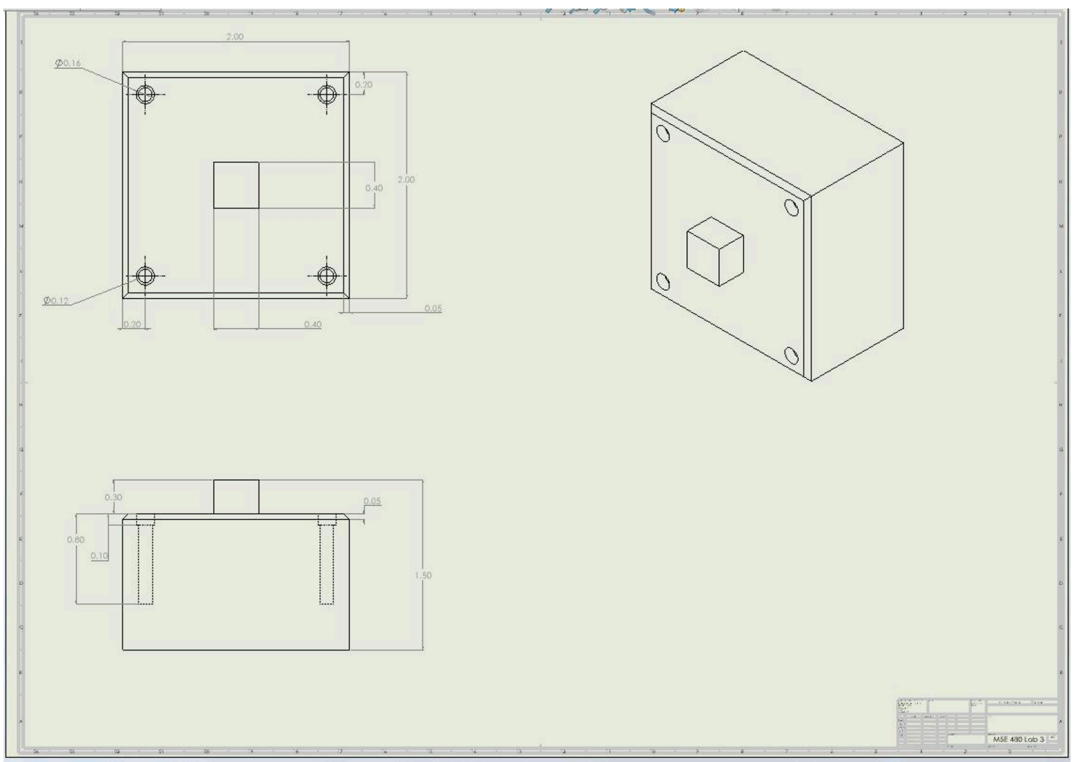

Figure 1: CAD Drawing of the Part

3. Milling Procedure

The following sections describe the step-by-step milling process starting from a 2" x 2" x 1.5" 6061 Aluminum rod.

3.1 Stock Preparation

- **Material Selection:** Select a 2" x 2" x 1.5" rod of 6061 Aluminum, a commonly used alloy known for its good machinability and strength.
- **Inspection:** Ensure the rod is clean, free of surface defects, and within dimensional tolerances to avoid issues during machining.

3.2 Fixturing

- **Setup:** Secure the aluminum rod in the milling machine's vise, with the 2" x 2" face oriented upward as the primary machining surface.
- **Alignment:** Verify that the rod is properly aligned and securely clamped to prevent movement or vibration during milling, which could compromise accuracy.

3.3 Tool Selection

- **Roughing Tool:** Use a 1/2" diameter, 4-flute carbide end mill. The carbide material and four-flute design provide durability and efficient material removal for aluminum.
- **Finishing Tool:** Use a 1/4" diameter, 2-flute carbide end mill. A smaller diameter and fewer flutes allow for finer cuts and better surface finishes.

3.4 Roughing Operation

- **Strategy:** Employ a 2D adaptive clearing toolpath to remove excess material efficiently. This method maintains constant tool engagement, reducing tool wear and machining time.
- **Parameters:**
  - Maximum stepdown (depth of cut): 0.1" per pass.
  - Feed rate: Adjusted based on spindle speed and tool specifications (e.g., approximately 50–100 inches per minute for aluminum).
  - Spindle speed: Around 8,000–10,000 RPM, suitable for roughing aluminum with a 1/2" end mill.
- **Stock Allowance:** Leave 0.01" of material on all surfaces for the finishing operation to ensure precision in the final dimensions.

3.5 Finishing Operation

- **Strategy:** Use a 2D contour or 3D parallel toolpath to finish the surfaces, depending on the part's geometry (e.g., flat surfaces or curved features).

- **Parameters:**
  - Stepover: 0.005" to minimize tool marks and achieve a smooth finish.
  - Depth of cut: Very light, approximately 0.001"–0.005" for the final pass.
  - Spindle speed: High, approximately 12,000 RPM, to enhance surface quality.
  - Feed rate: Low, around 20–40 inches per minute, to ensure the desired surface finish of 32 μinch (equivalent to ~0.8 μm Ra).
- **Technique:** Use climb milling to improve surface finish and extend tool life by reducing heat and friction.

3.6 Surface Finish Optimization

To achieve a surface finish of approximately 32 μinch:

- Use a sharp, coated carbide tool to minimize tool marks.
- Employ a high spindle speed and low feed rate to reduce vibration and improve smoothness.
- Apply coolant or lubricant during finishing to dissipate heat and enhance surface quality.
- Ensure the milling machine is rigid and free of vibrations to maintain precision.

4. Optimization for Producibility

The milling procedure is optimized as follows:

- **Minimum Waste Material:** The entire 2" x 2" x 1.5" stock is utilized, assuming the part fits within these dimensions, eliminating excess material waste.
- **Minimum Time Consumption:** Efficient roughing with adaptive clearing reduces material removal time, while performing all operations in a single setup minimizes setup changes and tool swaps.
- **Maximum Accuracy:** Precise fixturing and a small stock allowance for finishing ensure dimensional accuracy, with the final pass achieving the required tolerances and surface finish.

5. Additive Manufacturing Considerations

If the part were produced using additive manufacturing (e.g., Selective Laser Melting (SLM) or Direct Metal Laser Sintering (DMLS) for 6061 Aluminum), the following limitations would need to be considered:

- **Material Properties:**
  - Additively manufactured aluminum may exhibit anisotropy due to the layer-by-layer build process, potentially reducing strength or ductility

compared to wrought 6061 Aluminum.
    - Porosity or internal defects could affect mechanical performance unless mitigated by post-processing (e.g., hot isostatic pressing).
- **Surface Finish:**
    - Metal 3D printing typically results in surface roughness greater than 32 µinch (e.g., 100–200 µinch as-printed), necessitating post-processing such as machining or polishing to meet the required finish.
    - Additional steps increase production time and cost.
- **Dimensional Accuracy:**
    - Additive processes may struggle with tight tolerances or small features due to layer thickness and thermal distortion, often requiring secondary machining.
    - This reduces the advantage of additive manufacturing for precision parts.
- **Directional (Build Orientation):**
    - The part's orientation in the build chamber affects mechanical properties, surface finish, and the need for support structures.
    - Overhangs or complex geometries may require supports, which must be removed post-printing, adding labor and potential surface imperfections.
- **Time-Wise:**
    - For simple geometries like this part, additive manufacturing may be slower than milling when accounting for printing and post-processing time.
    - However, it could be faster for highly complex shapes with minimal material waste.
- **Cost:**
    - Additive manufacturing is generally more expensive for low-volume production due to high equipment and material costs, making milling more economical in this context.

In summary, while additive manufacturing offers design flexibility, traditional milling is preferable for this part due to superior material properties, surface finish, accuracy, and efficiency.

6. Fusion 360 Simulation

The milling process was simulated in Fusion 360 to verify the procedure:

1. **CAD Import:** The SolidWorks model of the part was imported into Fusion 360.
2. **Stock Definition:** A 2" x 2" x 1.5" stock was specified.
3. **Setup:** The workpiece orientation and vise fixturing were defined.
4. **Tool Selection:** The 1/2" roughing and 1/4" finishing end mills were selected from the tool library.
5. **Toolpaths:**

- Roughing: 2D adaptive clearing with specified parameters.
    - Finishing: 2D contour or 3D parallel with fine stepover and high spindle speed.
6. **Simulation:** The toolpaths were simulated to visualize material removal, check for collisions, and ensure the process meets objectives.
7. **G-Code:** G-code was generated for the CNC milling machine.

The simulation video, submitted separately, demonstrates the milling process from stock to finished part.

7. Conclusion

The outlined milling procedure efficiently machines the specified part from a 2" x 2" x 1.5" 6061 Aluminum rod, achieving a surface finish of approximately 32 µinch while maximizing producibility through minimal waste, reduced time, and high accuracy. The process leverages roughing and finishing operations tailored to aluminum, with parameters optimized for efficiency and quality. In contrast, additive manufacturing, while viable, introduces limitations in material properties, surface finish, and accuracy that make milling the preferred method for this application. The Fusion 360 simulation confirms the procedure's feasibility, aligning with the lab's objectives.